1

```
5412
5413  --1
5414  ANALYZE;
5415
5416  CREATE INDEX IF NOT EXISTS idx_flights_actual_departure
5417      ON flights (actual_departure);
5418
5419  --2
5420  SELECT flight_no, scheduled_departure, COUNT(*) AS duplicate_count
5421  FROM flights
5422  GROUP BY flight_no, scheduled_departure
5423  HAVING COUNT(*) > 1
```

Data Output | Messages | Notifications

```
CREATE INDEX

Query returned successfully in 508 msec.
```

✓ Query returned successfully in 508 msec.

Total rows: | Query complete 00:00:00.508 | CRLF | Ln 5417, Col 35

2

uery | Query History | Scratch Pad

```
433       ) AS rn
434      FROM flights
435  )
436  UPDATE flights AS f
437  SET flight_no = f.flight_no || '-DUP-' || f.flight_id::text
438  FROM ranked AS r
439  WHERE f.flight_id = r.flight_id
440    AND r.rn > 1;
441
442  SELECT flight_no, scheduled_departure, COUNT(*) AS duplicate_count
443  FROM flights
444  GROUP BY flight_no, scheduled_departure
445  HAVING COUNT(*) > 1;
446
447  CREATE UNIQUE INDEX IF NOT EXISTS ux_flights_flight_no_scheduled_departure
448      ON flights (flight_no, scheduled_departure);
449
450  --3
451  CREATE INDEX IF NOT EXISTS idx_flights_departure_arrival
452      ON flights (departure_airport_id, arrival_airport_id);
453
454  --4
```

ata Output | Messages | Notifications

```
REATE INDEX

uery returned successfully in 50 msec.
```

✓ Query returned successfully in 50 msec.

otal rows: | Query complete 00:00:00.050 | CRLF | Ln 5448, Col 49

3

```
--3
CREATE INDEX IF NOT EXISTS idx_flights_departure_arrival
    ON flights (departure_airport_id, arrival_airport_id);

--4
DROP INDEX IF EXISTS idx_flights_actual_departure;
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM flights
WHERE actual_departure = DATE '2023-04-28';
```

utput Messages Notifications

INDEX

returned successfully in 45 msec.

Query returned successfully in 45 msec.

ows: Query complete 00:00:00.045 CRLF Ln 5452, Col 59

4.

```
5454  --4
5455  DROP INDEX IF EXISTS idx_flights_actual_departure;
5456  EXPLAIN (ANALYZE, BUFFERS)
5457  SELECT *
5458  FROM flights
5459  WHERE actual_departure = DATE '2023-04-28';
5460
5461  CREATE INDEX idx_flights_actual_departure ON flights (actual_departure);
5462  ANALYZE flights;
5463  EXPLAIN (ANALYZE, BUFFERS)
5464  SELECT *
5465  FROM flights
5466  WHERE actual_departure = DATE '2023-04-28';
5467
5468  --5
5469  EXPLAIN (ANALYZE, BUFFERS)
5470  SELECT *
5471  FROM flights
5472  WHERE departure_airport_id = 12
5473    AND arrival_airport_id = 13;
5474
5475  --6
```

Data Output Messages Notifications

Showing rows: 1 to 12 Page No: 1 of 1

	QUERY PLAN text
1	Bitmap Heap Scan on flights (cost=4.31..13.70 rows=4 width=61) (actual time=0.349..0.351 rows=4.00 loops=1)
2	Recheck Cond: (actual_departure = '2023-04-28'::date)
3	Heap Blocks: exact=2
4	Buffers: shared hit=2 read=2
5	-> Bitmap Index Scan on idx_flights_actual_departure (cost=0.00..4.31 rows=4 width=0) (actual time=0.332..0.333 rows=4.00 loo...
6	Index Cond: (actual_departure = '2023-04-28'::date)
7	Index Searches: 1
8	Buffers: shared read=2
9	Planning:
10	Buffers: shared hit=51 read=1
11	Planning Time: 0.694 ms

Successfully run. Total query runtime: 96 msec. 12 rows affected.

Total rows: 12 Query complete 00:00:00.096 CRLF Ln 5466, Col 44

5

```
5467
5468  --5
5469  EXPLAIN (ANALYZE, BUFFERS)
5470  SELECT *
5471  FROM flights
5472  WHERE departure_airport_id = 12
5473    AND arrival_airport_id = 13;
5474
5475  --6
5476  CREATE UNIQUE INDEX IF NOT EXISTS ux_passengers_passport_number
5477      ON passengers (passport_number);
5478
5479  SELECT indexname, indexdef
5480  FROM pg_indexes
5481  WHERE schemaname = current_schema()
5482    AND tablename = 'passengers'
5483  ORDER BY indexname;
5484
5485  INSERT INTO passengers
```

Data Output | Messages | Notifications

Showing rows: 1 to 12 | Page No: 1 of 1

	QUERY PLAN text
1	Bitmap Heap Scan on flights (cost=4.31..12.03 rows=3 width=61) (actual time=0.046..0.047 rows=4.00 loops=1)
2	Recheck Cond: ((departure_airport_id = 12) AND (arrival_airport_id = 13))
3	Heap Blocks: exact=3
4	Buffers: shared hit=3 read=2
5	-> Bitmap Index Scan on idx_flights_departure_arrival (cost=0.00..4.31 rows=3 width=0) (actual time=0.037..0.037 rows=4.00 loo…
6	Index Cond: ((departure_airport_id = 12) AND (arrival_airport_id = 13))
7	Index Searches: 1
8	Buffers: shared read=2
9	Planning:
10	Buffers: shared hit=66
11	Planning Time: 1.685 ms

✓ Successfully run. Total query runtime: 69 msec. 12 rows affected.

Total rows: 12 | Query complete 00:00:00.069 | CRLF | Ln 5473, Col 31

6

Query | Query History | Scratch Pad

```
5490   'Kazakhstan', 'Kazakhstan', 'LAB7-PASS-201', CURRENT_DATE, CURRENT_DATE),
5491  (202, 'TestTwo', 'IndexTest', DATE '1998-09-22', 'Male',
5492   'Philippines', 'Philippines', 'LAB7-PASS-202', CURRENT_DATE, CURRENT_DATE);
5493
5494  SELECT passenger_id, first_name, last_name, passport_number
5495  FROM passengers
5496  WHERE passenger_id IN (201, 202);
5497
5498  --7
5499  CREATE INDEX IF NOT EXISTS idx_passengers_name_birth_citizenship
5500      ON passengers (first_name, last_name, date_of_birth, country_of_citizenship);
5501
5502  EXPLAIN (ANALYZE, BUFFERS)
5503  SELECT *
5504  FROM passengers
5505  WHERE country_of_citizenship = 'Philippines'
5506    AND date_of_birth >= DATE '1984-01-01'
5507    AND date_of_birth <  DATE '1985-01-01';
5508
5509  --8
5510  SELECT indexname, indexdef
```

Data Output | Messages | Notifications

Showing rows: 1 to 2 | Page No: 1 of 1

	passenger_id [PK] integer	first_name character varying (50)	last_name character varying (50)	passport_number character varying (50)
1	201	TestOne	IndexTest	LAB7-PASS-201
2	202	TestTwo	IndexTest	LAB7-PASS-202

✓ Successfully run. Total query runtime: 286 msec. 2 rows affected.

Total rows: 2 | Query complete 00:00:00.286 | CRLF | Ln 5496, Col 34

7

```
5496  WHERE passenger_id IN (201, 202);
5497
5498  --7
5499  CREATE INDEX IF NOT EXISTS idx_passengers_name_birth_citizenship
5500      ON passengers (first_name, last_name, date_of_birth, country_of_citizenship);
5501
5502  EXPLAIN (ANALYZE, BUFFERS)
5503  SELECT *
5504  FROM passengers
5505  WHERE country_of_citizenship = 'Philippines'
5506    AND date_of_birth >= DATE '1984-01-01'
5507    AND date_of_birth <  DATE '1985-01-01';
5508
5509  --8
5510  SELECT indexname, indexdef
```

Data Output | Messages | Notifications

Showing rows: 1 to 8 | Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on passengers (cost=0.00..6.54 rows=1 width=64) (actual time=0.018..0.036 rows=1.00 loops=1)
2	Filter: ((date_of_birth >= '1984-01-01'::date) AND (date_of_birth < '1985-01-01'::date) AND ((country_of_citizenship)::text = 'Philippines'...
3	Rows Removed by Filter: 201
4	Buffers: shared hit=3
5	Planning:
6	Buffers: shared hit=38 read=1
7	Planning Time: 2.307 ms
8	Execution Time: 0.052 ms

✓ Successfully run. Total query runtime: 79 msec. 8 rows affected.

Total rows: 8 | Query complete 00:00:00.079 | CRLF | Ln 5508, Col 1

8

```
5508
5509  --8
5510  SELECT indexname, indexdef
5511  FROM pg_indexes
5512  WHERE schemaname = current_schema()
5513    AND tablename = 'passengers'
5514  ORDER BY indexname;
5515
5516  DROP INDEX IF EXISTS idx_passengers_name_birth_citizenship;
5517  DROP INDEX IF EXISTS ux_passengers_passport_number;
5518
5519  --9
5520  EXPLAIN (ANALYZE, BUFFERS)
5521  SELECT *
5522  FROM booking
5523  WHERE passenger_id = 100;
5524
5525  CREATE INDEX IF NOT EXISTS idx_booking_passenger_id
5526      ON booking (passenger_id);
```

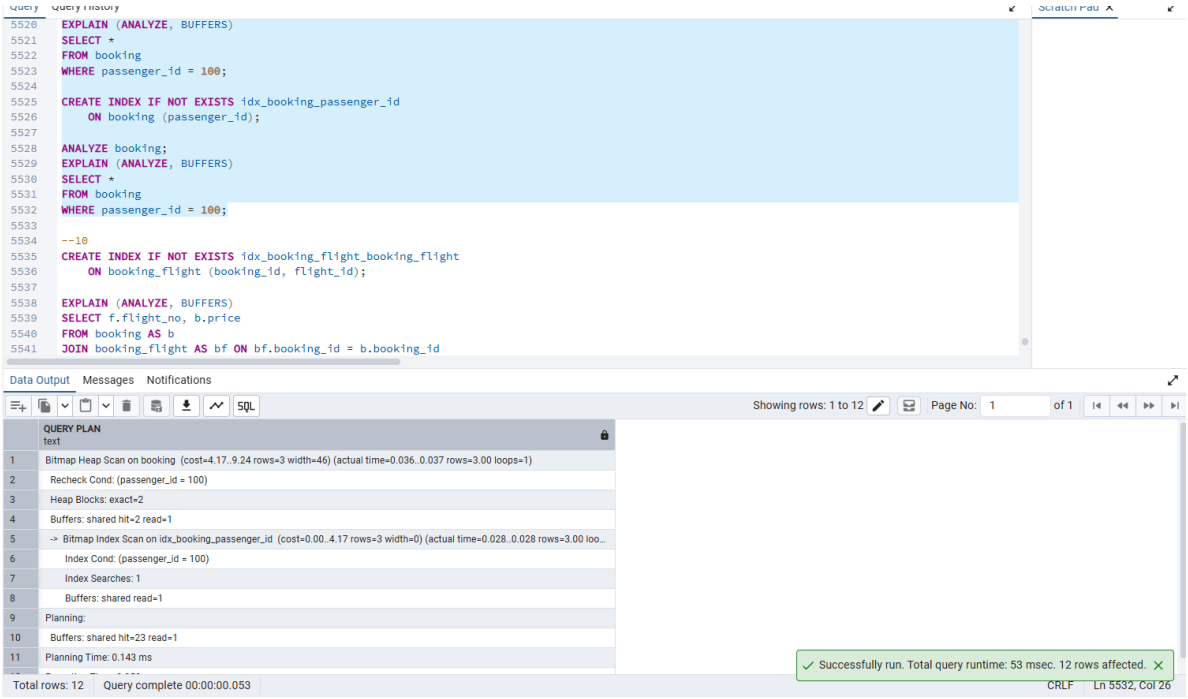
Data Output | Messages | Notifications

```
DROP INDEX

Query returned successfully in 67 msec.
```

✓ Query returned successfully in 67 msec.

9



```
5520  EXPLAIN (ANALYZE, BUFFERS)
5521  SELECT *
5522  FROM booking
5523  WHERE passenger_id = 100;
5524
5525  CREATE INDEX IF NOT EXISTS idx_booking_passenger_id
5526      ON booking (passenger_id);
5527
5528  ANALYZE booking;
5529  EXPLAIN (ANALYZE, BUFFERS)
5530  SELECT *
5531  FROM booking
5532  WHERE passenger_id = 100;
5533
5534  --10
5535  CREATE INDEX IF NOT EXISTS idx_booking_flight_booking_flight
5536      ON booking_flight (booking_id, flight_id);
5537
5538  EXPLAIN (ANALYZE, BUFFERS)
5539  SELECT f.flight_no, b.price
5540  FROM booking AS b
5541  JOIN booking_flight AS bf ON bf.booking_id = b.booking_id
```

Data Output | Messages | Notifications

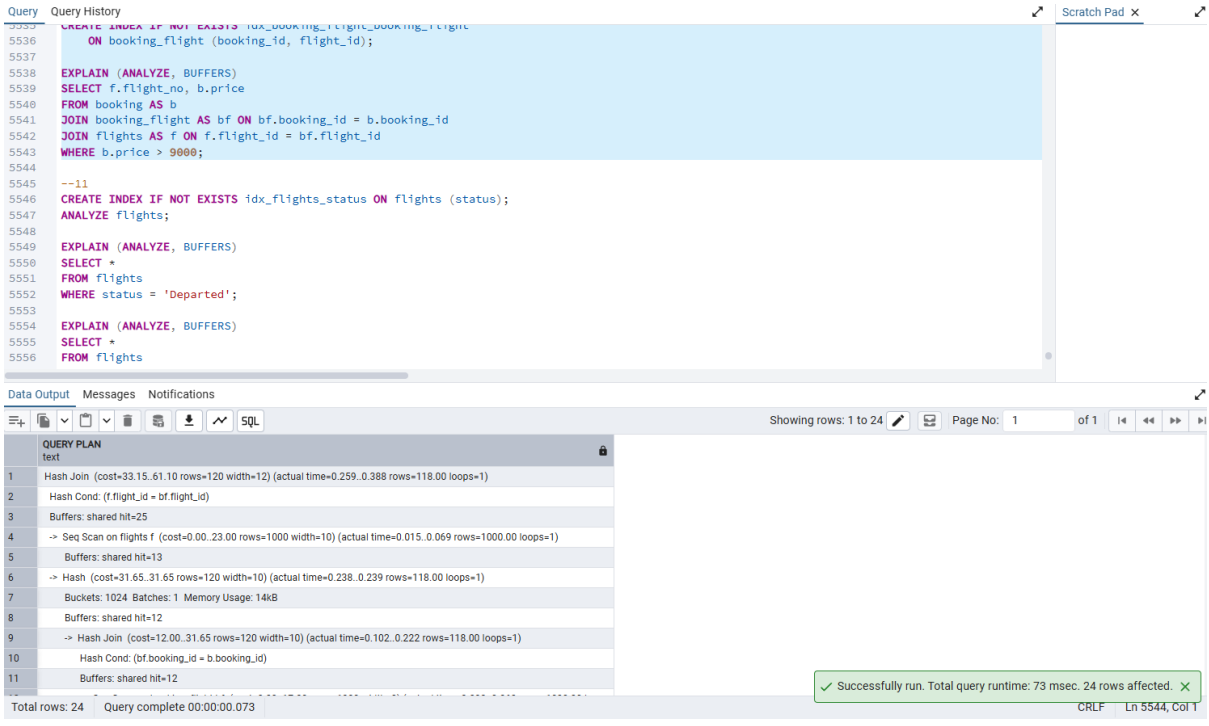
Showing rows: 1 to 12 | Page No: 1 of 1

	QUERY PLAN text
1	Bitmap Heap Scan on booking (cost=4.17..9.24 rows=3 width=46) (actual time=0.036..0.037 rows=3.00 loops=1)
2	Recheck Cond: (passenger_id = 100)
3	Heap Blocks: exact=2
4	Buffers: shared hit=2 read=1
5	-> Bitmap Index Scan on idx_booking_passenger_id (cost=0.00..4.17 rows=3 width=0) (actual time=0.028..0.028 rows=3.00 loo...
6	Index Cond: (passenger_id = 100)
7	Index Searches: 1
8	Buffers: shared read=1
9	Planning:
10	Buffers: shared hit=23 read=1
11	Planning Time: 0.143 ms

✓ Successfully run. Total query runtime: 53 msec. 12 rows affected.

Total rows: 12 | Query complete 00:00:00.053 | CRLF | Ln 5532, Col 26

10



Query | Query History | Scratch Pad

```
5536      ON booking_flight (booking_id, flight_id);
5537
5538  EXPLAIN (ANALYZE, BUFFERS)
5539  SELECT f.flight_no, b.price
5540  FROM booking AS b
5541  JOIN booking_flight AS bf ON bf.booking_id = b.booking_id
5542  JOIN flights AS f ON f.flight_id = bf.flight_id
5543  WHERE b.price > 9000;
5544
5545  --11
5546  CREATE INDEX IF NOT EXISTS idx_flights_status ON flights (status);
5547  ANALYZE flights;
5548
5549  EXPLAIN (ANALYZE, BUFFERS)
5550  SELECT *
5551  FROM flights
5552  WHERE status = 'Departed';
5553
5554  EXPLAIN (ANALYZE, BUFFERS)
5555  SELECT *
5556  FROM flights
```

Data Output | Messages | Notifications

Showing rows: 1 to 24 | Page No: 1 of 1

	QUERY PLAN text
1	Hash Join (cost=33.15..61.10 rows=120 width=12) (actual time=0.259..0.388 rows=118.00 loops=1)
2	Hash Cond: (f.flight_id = bf.flight_id)
3	Buffers: shared hit=25
4	-> Seq Scan on flights f (cost=0.00..23.00 rows=1000 width=10) (actual time=0.015..0.069 rows=1000.00 loops=1)
5	Buffers: shared hit=13
6	-> Hash (cost=31.65..31.65 rows=120 width=10) (actual time=0.238..0.239 rows=118.00 loops=1)
7	Buckets: 1024 Batches: 1 Memory Usage: 14kB
8	Buffers: shared hit=12
9	-> Hash Join (cost=12.00..31.65 rows=120 width=10) (actual time=0.102..0.222 rows=118.00 loops=1)
10	Hash Cond: (bf.booking_id = b.booking_id)
11	Buffers: shared hit=12

✓ Successfully run. Total query runtime: 73 msec. 24 rows affected.

Total rows: 24 | Query complete 00:00:00.073 | CRLF | Ln 5544, Col 1

11

```
5546  CREATE INDEX IF NOT EXISTS idx_flights_status ON flights (status);
5547  ANALYZE flights;
5548
5549  EXPLAIN (ANALYZE, BUFFERS)
5550  SELECT *
5551  FROM flights
5552  WHERE status = 'Departed';
5553
5554  EXPLAIN (ANALYZE, BUFFERS)
5555  SELECT *
5556  FROM flights
5557  WHERE status = 'Boarding';
5558
5559  --12
5560  CREATE INDEX IF NOT EXISTS idx_baggage_weight
5561      ON baggage (weight_in_kg);
5562  ANALYZE baggage;
5563
5564  EXPLAIN (ANALYZE, BUFFERS)
5565  SELECT *
5566  FROM baggage
```

Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on flights (cost=0.00..25.50 rows=500 width=61) (actual time=0.009..0.121 rows=500.00 loop...
2	Filter: ((status)::text = 'Boarding'::text)
3	Rows Removed by Filter: 500
4	Buffers: shared hit=13
5	Planning Time: 0.037 ms
6	Execution Time: 0.140 ms

✓ Successfully run. Total query runtime: 73 msec. 6 rows affected.

Total rows: 6 | Query complete 00:00:00.073 | CRLF | Ln 5557, Col 27

12

```
5566  FROM baggage
5567  WHERE weight_in_kg > 49;
5568
5569  EXPLAIN (ANALYZE, BUFFERS)
5570  SELECT *
5571  FROM baggage
5572  WHERE weight_in_kg > 5;
5573
5574  --13
5575  CREATE INDEX IF NOT EXISTS idx_airport_airport_name
5576      ON airport (airport_name);
5577
5578  CREATE INDEX IF NOT EXISTS idx_airport_name_pattern
5579      ON airport (airport_name varchar_pattern_ops);
5580
5581  ANALYZE airport;
5582  EXPLAIN (ANALYZE, BUFFERS)
5583  SELECT *
5584  FROM airport
5585  WHERE airport_name LIKE 'A%';
5586
5587  EXPLAIN (ANALYZE, BUFFERS)
```

Data Output | Messages | Notifications

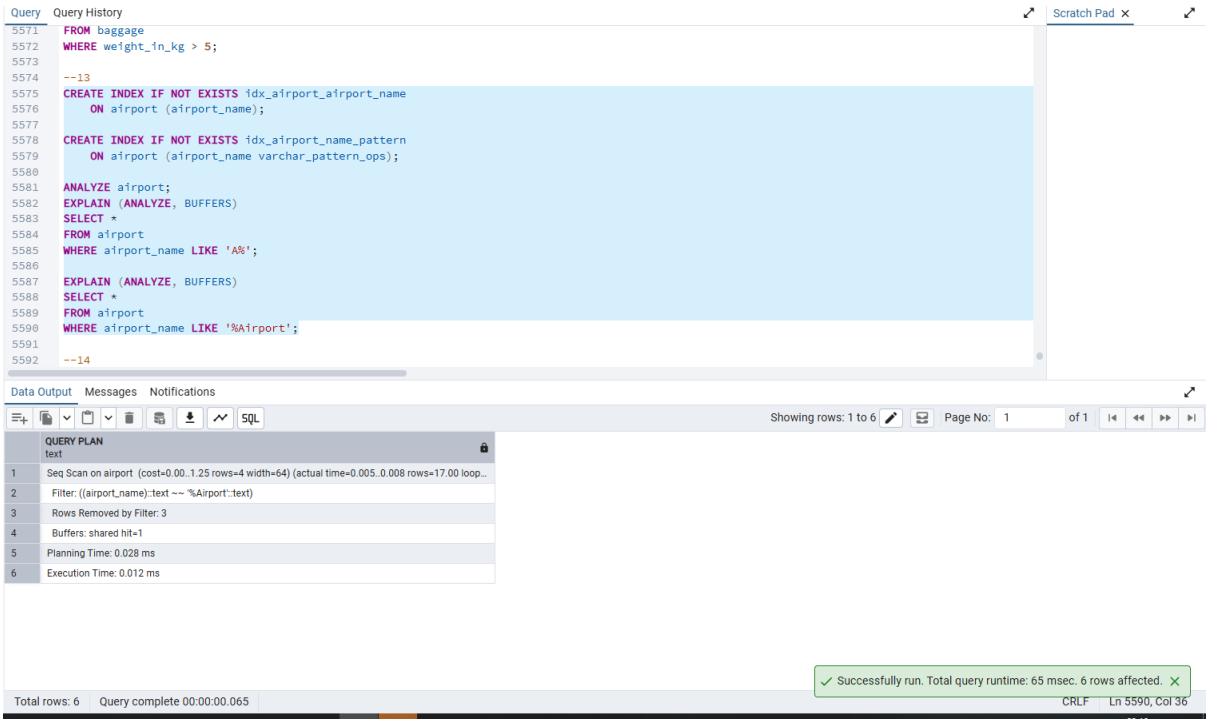
Showing rows: 1 to 6 | Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on baggage (cost=0.00..10.25 rows=451 width=22) (actual time=0.009..0.063 rows=451.00 loop...
2	Filter: (weight_in_kg > '5'::numeric)
3	Rows Removed by Filter: 49
4	Buffers: shared hit=4
5	Planning Time: 0.035 ms
6	Execution Time: 0.079 ms

✓ Successfully run. Total query runtime: 74 msec. 6 rows affected.

Total rows: 6 | Query complete 00:00:00.074 | CRLF | Ln 5573, Col 1

13



```
FROM baggage
WHERE weight_in_kg > 5;

--13
CREATE INDEX IF NOT EXISTS idx_airport_airport_name
    ON airport (airport_name);

CREATE INDEX IF NOT EXISTS idx_airport_name_pattern
    ON airport (airport_name varchar_pattern_ops);

ANALYZE airport;
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM airport
WHERE airport_name LIKE 'A%';

EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM airport
WHERE airport_name LIKE '%Airport';

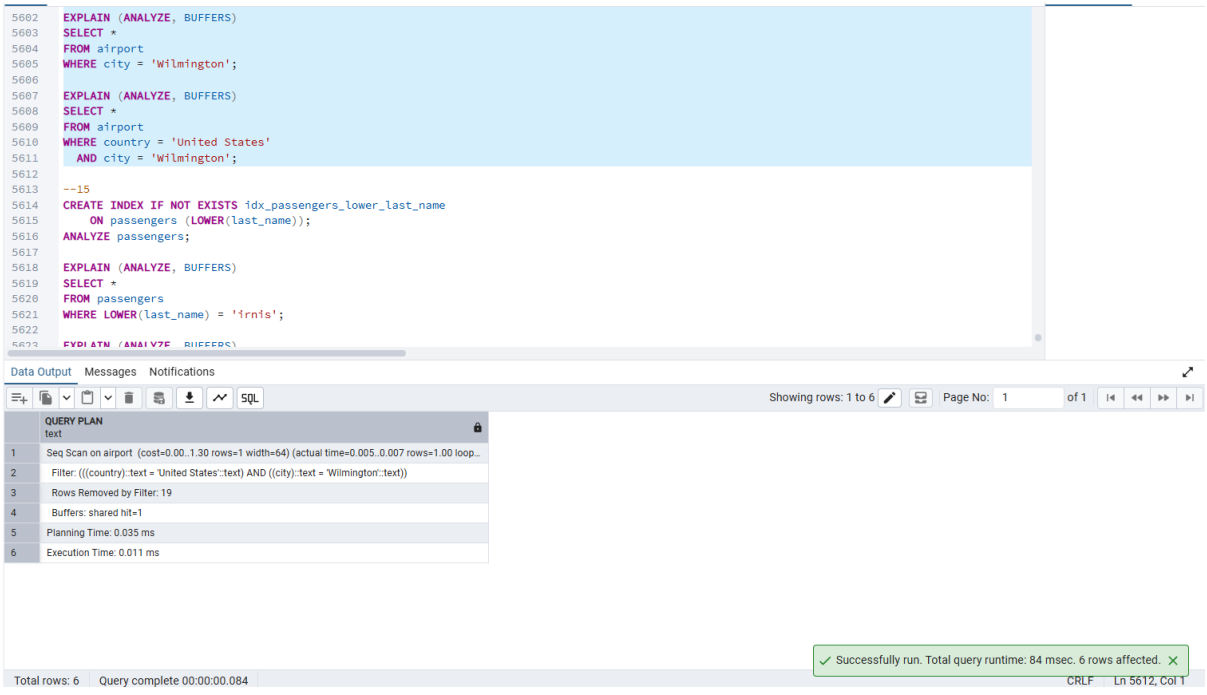
--14
```

QUERY PLAN text
Seq Scan on airport (cost=0.00..1.25 rows=4 width=64) (actual time=0.005..0.008 rows=17.00 loop...
Filter: ((airport_name)::text ~~ '%Airport'::text)
Rows Removed by Filter: 3
Buffers: shared hit=1
Planning Time: 0.028 ms
Execution Time: 0.012 ms

Successfully run. Total query runtime: 65 msec. 6 rows affected.

Total rows: 6 Query complete 00:00:00.065 CRLF Ln 5590, Col 36

14



```
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM airport
WHERE city = 'Wilmington';

EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM airport
WHERE country = 'United States'
  AND city = 'Wilmington';

--15
CREATE INDEX IF NOT EXISTS idx_passengers_lower_last_name
    ON passengers (LOWER(last_name));
ANALYZE passengers;

EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM passengers
WHERE LOWER(last_name) = 'irnis';
```

QUERY PLAN text
Seq Scan on airport (cost=0.00..1.30 rows=1 width=64) (actual time=0.005..0.007 rows=1.00 loop...
Filter: (((country)::text = 'United States'::text) AND ((city)::text = 'Wilmington'::text))
Rows Removed by Filter: 19
Buffers: shared hit=1
Planning Time: 0.035 ms
Execution Time: 0.011 ms

Successfully run. Total query runtime: 84 msec. 6 rows affected.

Total rows: 6 Query complete 00:00:00.084 CRLF Ln 5612, Col 1

15

```
5617
5618  EXPLAIN (ANALYZE, BUFFERS)
5619  SELECT *
5620  FROM passengers
5621  WHERE LOWER(last_name) = 'irnis';
5622
5623  EXPLAIN (ANALYZE, BUFFERS)
5624  SELECT *
5625  FROM passengers
5626  WHERE last_name = 'Irnis';
5627
5628  --16
5629  CREATE INDEX IF NOT EXISTS idx_booking_passenger_price
5630      ON booking (passenger_id) INCLUDE (price);
5631  ANALYZE booking;
5632
5633  EXPLAIN (ANALYZE, BUFFERS)
5634  SELECT price
5635  FROM booking
5636  WHERE passenger_id = 100;
5637
5638  --17
```

Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on passengers (cost=0.00..5.53 rows=1 width=64) (actual time=0.007..0.021 rows=1.00 loop...
2	Filter: ((last_name)::text = 'Irnis'::text)
3	Rows Removed by Filter: 201
4	Buffers: shared hit=3
5	Planning Time: 0.032 ms
6	Execution Time: 0.026 ms

✓ Successfully run. Total query runtime: 73 msec. 6 rows affected.

Total rows: 6 | Query complete 00:00:00.073 | CRLF | Ln 5627, Col 1

16

Query | Query History | Scratch Pad

```
5622
5623  EXPLAIN (ANALYZE, BUFFERS)
5624  SELECT *
5625  FROM passengers
5626  WHERE last_name = 'Irnis';
5627
5628  --16
5629  CREATE INDEX IF NOT EXISTS idx_booking_passenger_price
5630      ON booking (passenger_id) INCLUDE (price);
5631  ANALYZE booking;
5632
5633  EXPLAIN (ANALYZE, BUFFERS)
5634  SELECT price
5635  FROM booking
5636  WHERE passenger_id = 100;
5637
5638  --17
5639  SELECT airline_code, COUNT(*) AS duplicate_count
5640  FROM airline
5641  WHERE airline_code IS NOT NULL
5642  GROUP BY airline_code
5643  HAVING COUNT(*) > 1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 12 | Page No: 1 of 1

	QUERY PLAN text
1	Bitmap Heap Scan on booking (cost=4.17..9.24 rows=3 width=6) (actual time=0.018..0.020 rows=3.00 loops=1)
2	Recheck Cond: (passenger_id = 100)
3	Heap Blocks: exact=2
4	Buffers: shared hit=3
5	-> Bitmap Index Scan on idx_booking_passenger_id (cost=0.00..4.17 rows=3 width=0) (actual time=0.009..0.009 rows=3.00 loo...
6	Index Cond: (passenger_id = 100)
7	Index Searches: 1
8	Buffers: shared hit=1
9	Planning:
10	Buffers: shared hit=9 read=1
11	Planning Time: 0.125 ms

✓ Successfully run. Total query runtime: 66 msec. 12 rows affected.

Total rows: 12 | Query complete 00:00:00.066 | CRLF | Ln 5636, Col 26

17

```
5637
5638  --17
5639  SELECT airline_code, COUNT(*) AS duplicate_count
5640  FROM airline
5641  WHERE airline_code IS NOT NULL
5642  GROUP BY airline_code
5643  HAVING COUNT(*) > 1;
5644
5645  CREATE UNIQUE INDEX IF NOT EXISTS ux_airline_airline_code
5646      ON airline (airline_code);
5647
5648  SELECT indexname, indexdef
5649  FROM pg_indexes
5650  WHERE schemaname = current_schema()
5651    AND tablename = 'airline'
5652  ORDER BY indexname;
5653
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	indexname name	indexdef text
1	airline_pkey	CREATE UNIQUE INDEX airline_pkey ON public.airline USING btree (airline_id)
2	ux_airline_airline_co…	CREATE UNIQUE INDEX ux_airline_airline_code ON public.airline USING btree (airline_c…

Successfully run. Total query runtime: 82 msec. 2 rows affected.

Total rows: 2 Query complete 00:00:00.082 CRLF Ln 5653, Col 1